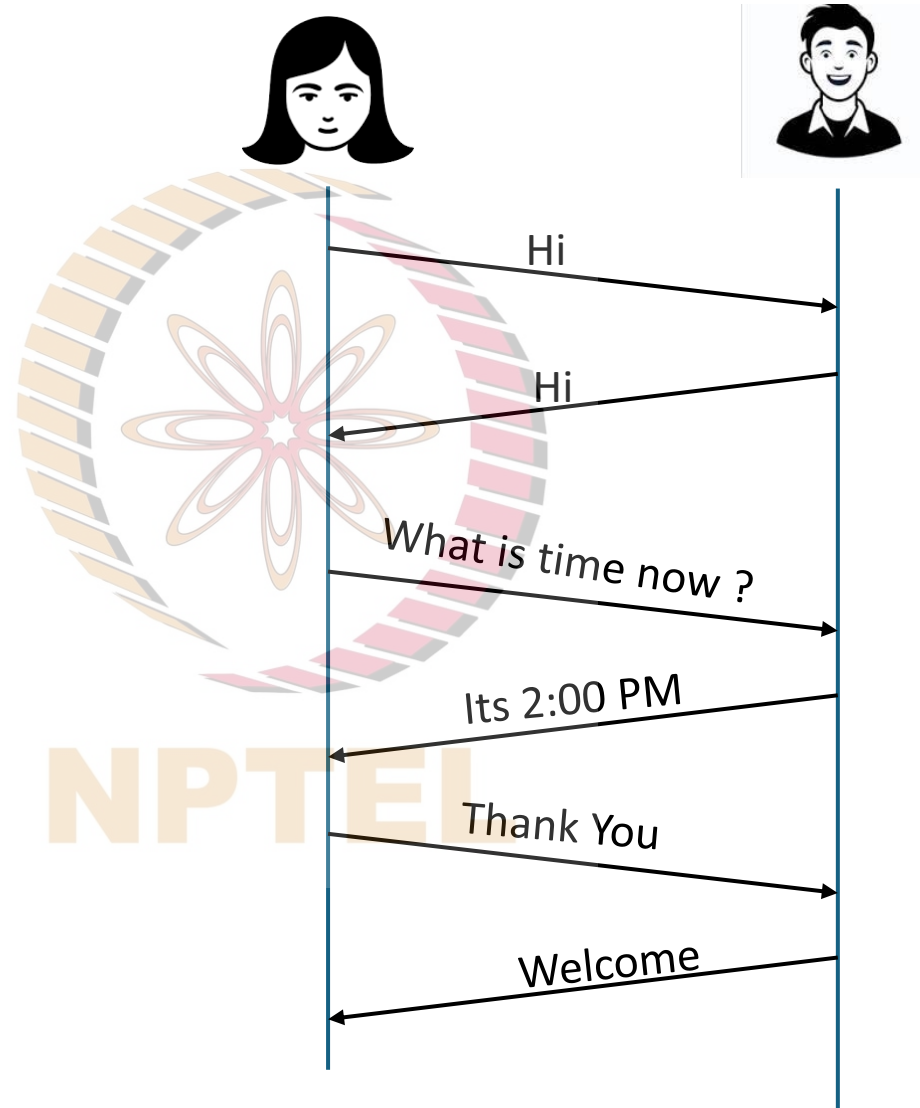


6. How to Study Networks ?

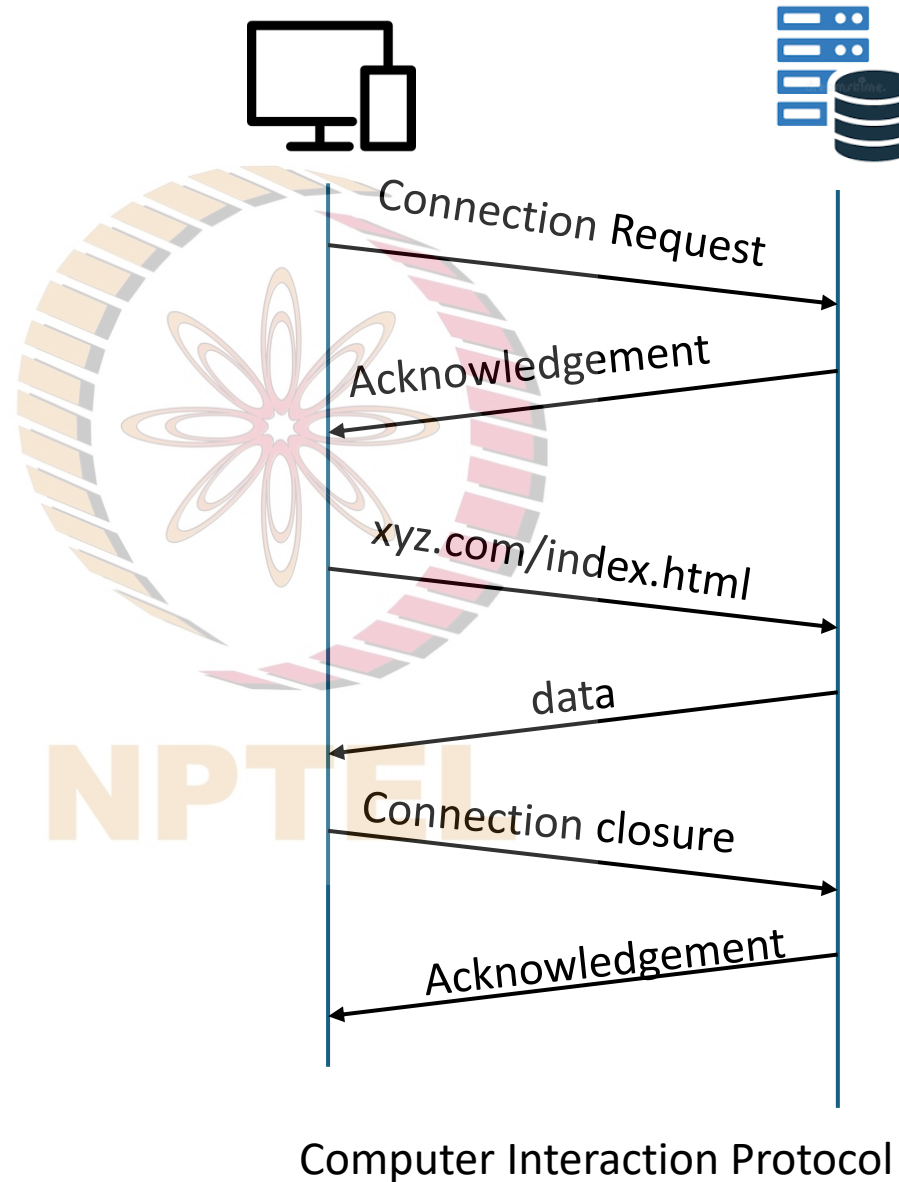
- ❑ Study Protocols
- ❑ What is Protocol ?



Human Interaction Protocol

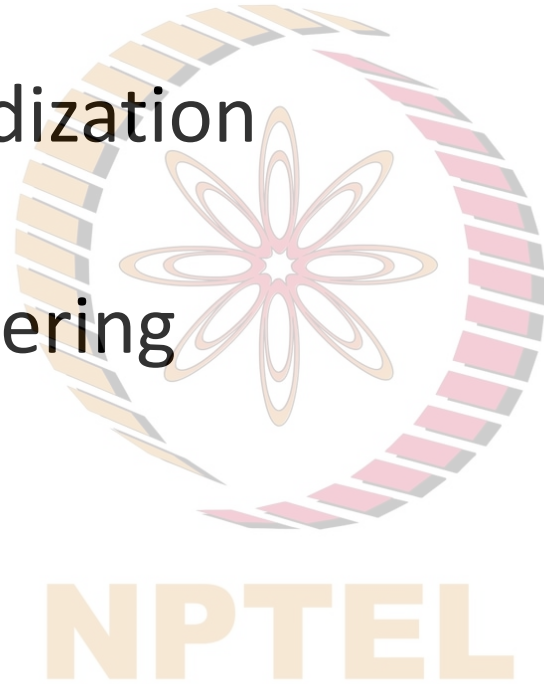
6. How to Study Networks ?

- ☐ Study Protocols
- ☐ What is Protocol ?



6. How to Study Networks: Standardization of Network Functionalities

- ❑ 1977 ISO started standardization process
- ❑ 1984 First standard of layering architecture was released



Application
Presentation
Session
Transport
Network
Data Link
Physical

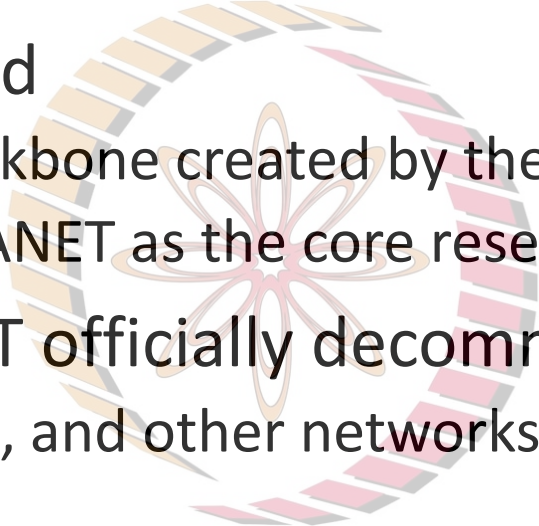
OSI Reference Model

ARPANET Continues to Evolve

- ❑ ARPANET was using Network Control Protocol (NCP)
- ❑ It was more monolithic Handling file transfers, establishing connections
- ❑ Working for Uniform network structure
- ❑ Diversity became issue
- ❑ 1973 - BBN and UCLA — **Vint Cerf** and **Bob Kahn** proposed new protocol and authored a paper “*A Protocol for Packet Network Intercommunication*”
- ❑ 1978 – Split into two protocols TCP and IP
- ❑ 1983 - ARPANET switched to TCP/IP

ARPANET Closure

- ❑ 1986 — NSFNET launched
 - ❑ High-speed academic backbone created by the National Science Foundation
 - ❑ Begins to supersede ARPANET as the core research network
- ❑ Feb 28 1990 — ARPANET officially decommissioned
 - ❑ NSFNET, commercial ISPs, and other networks take over

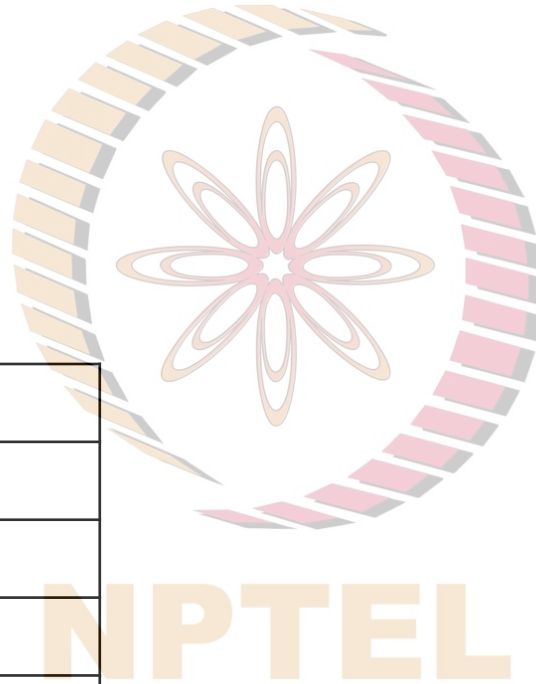
The NPTEL logo is a circular emblem. It features a stylized, multi-petaled flower or star shape in the center, rendered in a reddish-pink color. Surrounding this central motif is a circular border composed of alternating orange and pink segments, giving it a segmented or orbital appearance.

NPTEL

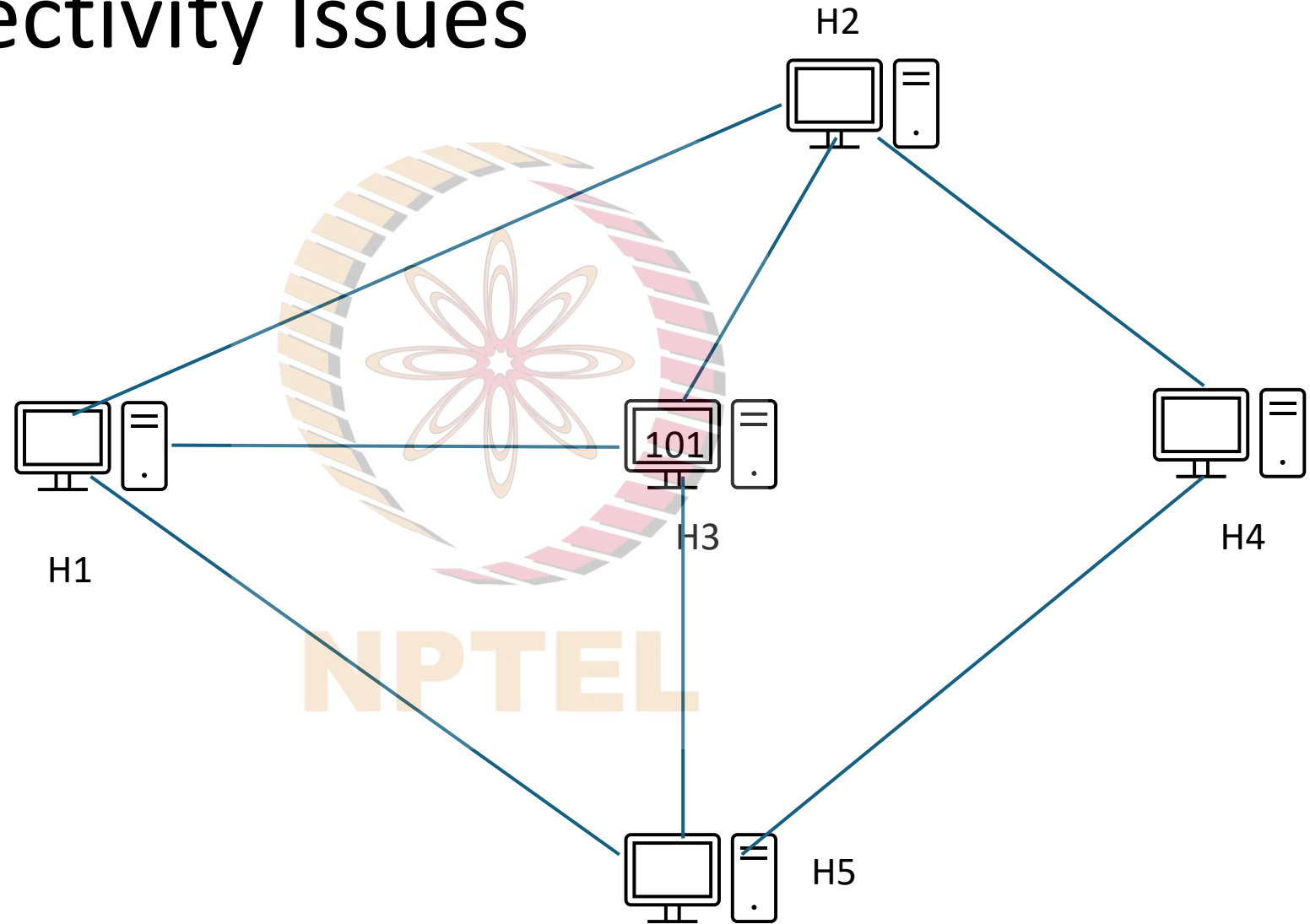
6. How to Study Networks ?

Study Protocols

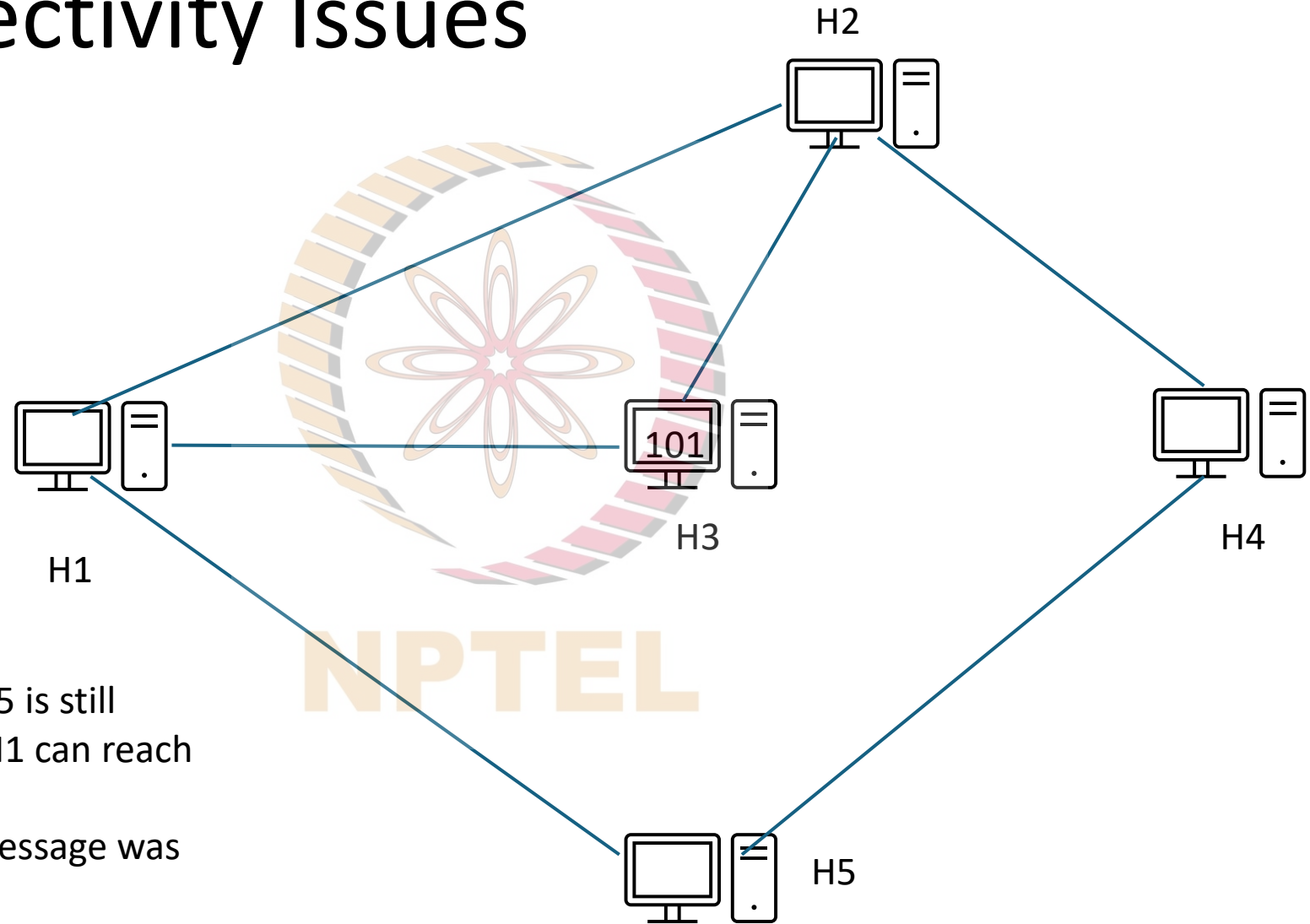
Application
Presentation
Session
Transport
Network
Data Link
Physical



Interconnectivity Issues



Interconnectivity Issues



1. How H3 Understands that H5 is still connected to network and H1 can reach H5 ?
2. How H3 understands that message was indeed delivered ?
3. Did the message reach as H3 sent ?

History of RFCs

- ❑ The first RFC, "Host Software," was published by Steve Crocker

Network Working Group
Request for Comments: 1

Steve Crocker
UCLA
7 April 1969

Title: Host Software
Author: Steve Crocker
Installation: UCLA
Date: 7 April 1969

Network Working Group Request for Comment: 1

CONTENTS

INTRODUCTION

I. A Summary of the IMP Software

Messages

Links

IMP Transmission and Error Checking

Open Questions on the IMP Software

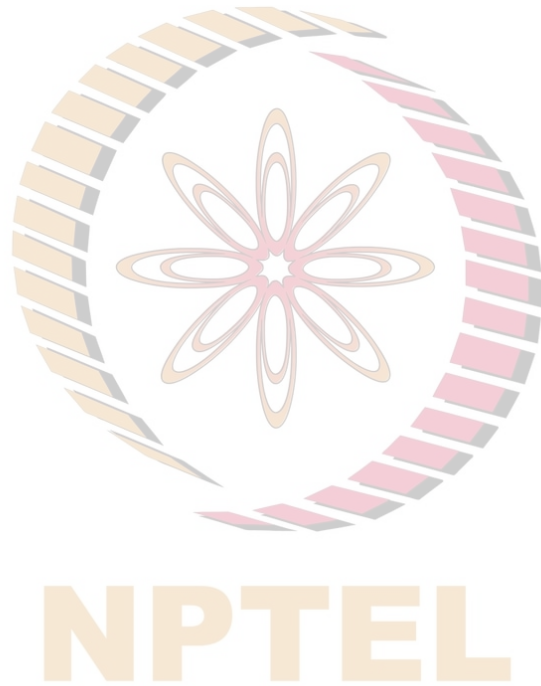
II. Some Requirements Upon the Host-to-Host Software

Simple Use

NPTTEL

History of RFCs

- ❑ Types of RFCs
 - ❑ Standards
 - ❑ Experimental
 - ❑ Informational
 - ❑ Historical



RFC Count

RFC 9904	HTML , TEXT , PDF , XML	DNSSEC Cryptographic Algorithm Recommendation Update Process	W. Hardaker, W. Kumari	November 2025	Obsoletes RFC 8624 , Updates RFC 9157	Proposed Standard
RFC 9905	HTML , TEXT , PDF , XML	Deprecating the Use of SHA-1 in DNSSEC Signature Algorithms	W. Hardaker, W. Kumari	November 2025	Updates RFC 4034 , RFC 5155	Proposed Standard
RFC 9906	HTML , TEXT , PDF , XML	Deprecate Usage of ECC-GOST within DNSSEC	W. Hardaker, W. Kumari	November 2025		Proposed Standard
RFC 9900	HTML , TEXT , PDF , XML	Updates to NETCONF Transport Port Numbers	M. Boucadair	December 2025		Proposed Standard
RFC 9887	HTML , TEXT , PDF , XML	Terminal Access Controller Access-Control System Plus (TACACS+) over TLS 1.3	T. Dahm, J. Heasley, D.C. Medway Gash, A. Ota	December 2025	Updates RFC 8907	Proposed Standard
RFC 9902	HTML , TEXT , PDF , XML	A YANG Data Model for IS-IS Segment Routing over the MPLS Data Plane	S. Litkowski, Y. Qu, A. Lindem, I. Chen, J. Tantsura	December 2025		Proposed Standard
RFC 9903	HTML , TEXT , PDF , XML	A YANG Data Model for OSPF Segment Routing over the MPLS Data Plane	Y. Qu, A. Lindem, J. Zhang, I. Chen	December 2025		Proposed Standard
RFC 9937	HTML , TEXT , PDF , XML	Proportional Rate Reduction (PRR)	M. Mathis, N. Cardwell, Y. Cheng, N. Dukkapati	December 2025	Obsoletes RFC 6937	Proposed Standard
RFC 9899	HTML , TEXT , PDF , XML	Extensions to the YANG Data Model for Access Control Lists (ACLs)	O. Gonzalez de Dios, S. Barguil, M. Boucadair, Q. Wu	December 2025		Proposed Standard
RFC 9911	HTML , TEXT , PDF , XML	Common YANG Data Types	J. Schönwälder, Ed.	December 2025	Obsoletes RFC 6991	Proposed Standard
RFC 9909	HTML , TEXT , PDF , XML	Internet X.509 Public Key Infrastructure -- Algorithm Identifiers for the Stateless Hash-Based Digital Signature Algorithm (SLH-DSA)	K. Bashiri, S. Fluhrer, S. Gazdag, D. Van Geest, S. Kousidis	December 2025		Proposed Standard
RFC 9886	HTML , TEXT , PDF , XML , HTML with inline errata	DRIP Entity Tags (DETs) in the Domain Name System	A. Wiethuechter, Ed., J. Reid	December 2025	Errata	Proposed Standard
RFC 9847	HTML , TEXT , PDF , XML	IANA Registry Updates for TLS and DTLS	J. Salowey, S. Turner	December 2025	Updates RFC 8447	Proposed Standard